A Self-Contained Derivation of SER-relax

1 Model

Let $x \in \mathbb{R}^J$ be the vector of marginal association statistics and let N be the GWAS sample size. Let $\bar{R} \in \mathbb{R}^{J \times J}$ be a positive definite reference LD correlation matrix, and write

$$\Omega = \bar{R}^{-1}.$$

For each candidate active coordinate j, define the reference conditional mean and Schur complement

$$\mu_j = \bar{R}_{-j,j}, \qquad S_j = \bar{R}_{-j,-j} - \bar{R}_{-j,j}\bar{R}_{j,-j}.$$

Because $\bar{R}$ is a correlation matrix, $\bar{R}_{jj} = 1$. For $z_j \in \mathbb{R}^{J-1}$, let $u_j(z_j) \in \mathbb{R}^J$ denote the vector whose j-th entry is 1 and whose remaining entries are z_j.

SER-relax is a single-effect model that keeps the observation covariance fixed at $\bar{R}/N$, but allows the active LD column to move around the reference LD column. The model is

$$\gamma \sim \mathrm{Unif}\{1, \dots, J\}, \tag{1}$$

$$\beta \sim \mathcal{N}(0, \sigma_0^2), \tag{2}$$

$$z_j \mid \gamma = j, \nu_0 \sim \mathcal{N}_{J-1}(\mu_j, \tau^2 S_j), \tag{3}$$

$$x \mid \beta, \gamma = j, z_j \sim \mathcal{N}_J\left(\beta u_j(z_j), \frac{\bar{R}}{N}\right), \tag{4}$$

where

$$\tau^2 = \frac{1}{\nu_0 + 3}, \qquad \tau^{-2} = \nu_0 + 3.$$

Large ν_0 makes z_j concentrate near the reference column μ_j. Small ν_0 allows more LD-column relaxation.

2 Useful Block Identities

The block inverse formula for $\bar{R}$ gives

$$\Omega_{-j,-j} = S_j^{-1}, \qquad \Omega_{-j,j} = -S_j^{-1}\mu_j.$$

Therefore, for any z_j, with

$$\delta_j = x_{-j} - \mu_j x_j, \qquad y_j = z_j - \mu_j,$$

we have

$$u_j(z_j)^\top \Omega u_j(z_j) = 1 + y_j^\top S_j^{-1} y_j, \tag{5}$$

$$u_j(z_j)^\top \Omega x = x_j + y_j^\top S_j^{-1} \delta_j. \tag{6}$$

These two identities are the algebraic core of the CAVI updates.

3 Variational Family

Use the mean-field approximation

$$q(\gamma = j, \beta, z_j) = \pi_j q_j(\beta) q_j(z_j),$$

where

$$q_j(\beta) = \mathcal{N}(m_{\beta j}, v_{\beta j}), \qquad q_j(z_j) = \mathcal{N}_{J-1}(m_{zj}, V_{zj}).$$

Write

$$\bar{\beta}_j = m_{\beta j}, \qquad \overline{\beta^2}_j = v_{\beta j} + m_{\beta j}^2.$$

For the current $q_j(z_j)$, define

$$Q_j = \operatorname{tr}(S_j^{-1} V_{zj}) + (m_{zj} - \mu_j)^\top S_j^{-1} (m_{zj} - \mu_j), \tag{7}$$

$$A_j = 1 + Q_j, \tag{8}$$

$$B_j = x_j + (m_{zj} - \mu_j)^\top S_j^{-1} \delta_j. \tag{9}$$

Here $Q_j = \mathbb{E}_j[y_j^\top S_j^{-1} y_j]$, $A_j = \mathbb{E}_j[u_j(z_j)^\top \Omega u_j(z_j)]$, and $B_j = \mathbb{E}_j[u_j(z_j)^\top \Omega x]$.

4 CAVI Derivation

Conditional on $\gamma = j$, the log joint density, up to constants that do not depend on β or z_j, is

$$\begin{aligned}\log p(x, \beta, z_j, \gamma = j) = \log(1/J) &- \frac{N}{2}\{x - \beta u_j(z_j)\}^\top \Omega \{x - \beta u_j(z_j)\} \\ &- \frac{\beta^2}{2\sigma_0^2} - \frac{\tau^{-2}}{2}(z_j - \mu_j)^\top S_j^{-1} (z_j - \mu_j).\end{aligned} \tag{10}$$

4.1 Update for $q_j(\beta)$

The optimal coordinate update satisfies

$$\log q_j^*(\beta) = \mathbb{E}_{q_j(z_j)} \log p(x, \beta, z_j, \gamma = j) + \text{const}.$$

Keeping only the terms that depend on β, using (5)–(6),

$$\log q_j^*(\beta) = N\beta B_j - \frac{1}{2}\left(\sigma_0^{-2} + N A_j\right)\beta^2 + \text{const}. \tag{11}$$

Thus the update is Gaussian:

$$v_{\beta j} = \left(\sigma_0^{-2} + N A_j\right)^{-1}, \tag{12}$$

$$m_{\beta j} = v_{\beta j} N B_j. \tag{13}$$

4.2 Update for $q_j(z_j)$

The optimal relaxed-column update satisfies

$$\log q_j^*(z_j) = \mathbb{E}_{q_j(\beta)} \log p(x, \beta, z_j, \gamma = j) + \text{const.}$$

Let $y_j = z_j - \mu_j$. The terms depending on y_j are

$$\log q_j^*(z_j) = N\bar{\beta}_j y_j^\top S_j^{-1}\delta_j - \frac{1}{2}\left(N\overline{\beta^2}_j + \tau^{-2}\right) y_j^\top S_j^{-1} y_j + \text{const.} \tag{14}$$

Completing the square gives

$$V_{zj} = \frac{S_j}{N\overline{\beta^2}_j + \tau^{-2}}, \tag{15}$$

$$m_{zj} = \mu_j + \frac{N\bar{\beta}_j}{N\overline{\beta^2}_j + \tau^{-2}}\delta_j. \tag{16}$$

4.3 Update for π_j

The optimal categorical probabilities are proportional to the local variational lower bounds:

$$\pi_j = \frac{\exp(\mathcal{L}_j)}{\sum_{k=1}^J \exp(\mathcal{L}_k)}.$$

For SER-relax,

$$\begin{aligned} \mathcal{L}_j = {} & \log(1/J) + \mathbb{E}_j \log p(x \mid \beta, z_j, \gamma = j) + \mathbb{E}_j \log p(\beta) + \mathbb{E}_j \log p(z_j \mid \nu_0) \\ & - \mathbb{E}_j \log q_j(\beta) - \mathbb{E}_j \log q_j(z_j). \end{aligned} \tag{17}$$

The expected log likelihood is

$$\begin{aligned} \mathbb{E}_j \log p(x \mid \beta, z_j, \gamma = j) = {} & -\frac{J}{2}\log(2\pi) - \frac{1}{2}\log\left|\frac{\bar{R}}{N}\right| \\ & - \frac{N}{2}\left[x^\top \Omega x - 2\bar{\beta}_j B_j + \overline{\beta^2}_j A_j\right]. \end{aligned} \tag{18}$$

The Gaussian prior and entropy terms are

$$\mathbb{E}_j \log p(\beta) = -\frac{1}{2}\log(2\pi\sigma_0^2) - \frac{\overline{\beta^2}_j}{2\sigma_0^2}, \tag{19}$$

$$\mathbb{E}_j \log p(z_j \mid \nu_0) = -\frac{J-1}{2}\log(2\pi) - \frac{1}{2}\log|S_j| + \frac{J-1}{2}\log\tau^{-2} - \frac{\tau^{-2}}{2}Q_j, \tag{20}$$

$$-\mathbb{E}_j \log q_j(\beta) = \frac{1}{2}\log(2\pi e\, v_{\beta j}), \tag{21}$$

$$-\mathbb{E}_j \log q_j(z_j) = \frac{1}{2}\log\left\{(2\pi e)^{J-1}|V_{zj}|\right\}. \tag{22}$$

These expressions define a numerically stable coordinate update by computing $\mathcal{L}_j$ for all j and applying a softmax after subtracting $\max_j \mathcal{L}_j$.

5 Equivalent Scalar Form

For implementation, one can avoid explicitly forming each S_j^{-1}. For any vector $r \in \mathbb{R}^J$, define

$$C_j(r) = (r_{-j} - \mu_j r_j)^\top S_j^{-1}(r_{-j} - \mu_j r_j).$$

The block inverse identity implies

$$C_j(r) = r^\top \Omega r - r_j^2.$$

When $r = x$, set

$$z_{\text{denom},j} = N\overline{\beta^2}_j + \tau^{-2}, \qquad z_{\text{scale},j} = \frac{N\bar{\beta}_j}{z_{\text{denom},j}}.$$

After the $q_j(z_j)$ update,

$$Q_j = \frac{J-1}{z_{\text{denom},j}} + z_{\text{scale},j}^2 C_j(x), \tag{23}$$

$$A_j = 1 + Q_j, \tag{24}$$

$$B_j = x_j + z_{\text{scale},j} C_j(x). \tag{25}$$

This representation reduces the coordinate-specific calculations to scalar operations once $x^\top \Omega x$ is available.

6 Optional Empirical Bayes Update for ν_0

Holding the variational distribution fixed, the ν_0-dependent part of the SER-relax ELBO is, up to constants,

$$M(\nu_0) = \sum_{j=1}^{J} \pi_j \left\{ \frac{J-1}{2} \log \tau^{-2} - \frac{\tau^{-2}}{2} Q_j \right\}.$$

Because $\sum_j \pi_j = 1$, maximizing over τ^2 gives

$$\hat{\tau}^2 = \frac{\sum_{j=1}^J \pi_j Q_j}{J-1}, \qquad \hat{\nu}_0 = \frac{1}{\hat{\tau}^2} - 3.$$

In practice ν_0 should be constrained to be positive and bounded away from numerically extreme values.